

RYNGE, Sweden

WMO: 026090

Lat: 55.4742N

Lon: 13.6683E

Elev: 50

StdP: 100.73

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.8	-7.4	-12.9	1.2	-8.5	-10.6	1.5	-6.1	13.9	4.0	13.0	3.9	2.6	0	0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	25.4	18.4	23.9	17.6	22.5	17.1	19.7	23.4	18.7	22.0	17.9	21.1	4.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.2	21.5	17.4	12.5	20.3	16.6	11.9	19.5	56.5	23.5	53.2	22.2	50.8	21.2	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	10.0	8.9	-13.7	28.0	3.9	1.6	-16.5	29.1	-18.8	30.1	-21.0	31.0	-23.8	32.2		
(5)			-14.0	21.1	3.9	1.2	-16.8	21.9	-19.1	22.6	-21.3	23.3	-24.1	24.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.2	0.0	0.3	2.6	6.8	10.8	14.4	17.1	16.9	13.2	9.1	5.4	1.8
	DBStd	6.82	3.93	3.46	3.53	2.92	2.93	2.46	2.56	2.42	2.50	3.21	3.31	4.07
	HDD10.0	1392	309	272	231	103	24	1	0	0	3	54	140	254
	HDD18.3	3715	567	506	489	347	233	121	55	58	154	285	388	512
	CDD10.0	753	0	0	0	7	49	132	222	214	100	28	2	0
	CDD18.3	34	0	0	0	0	0	2	18	13	1	0	0	0
	CDH23.3	185	0	0	0	0	1	21	117	47	0	0	0	0
	CDH26.7	12	0	0	0	0	0	1	8	3	0	0	0	0
Wind	WSAvg	4.3	5.0	5.1	4.6	4.2	4.2	3.8	3.3	3.7	4.0	4.1	4.6	4.8
Precipitation	PrecAvg	675	54	46	40	35	47	56	62	73	66	65	67	64
	PrecMax	878	120	97	95	66	106	132	182	216	151	128	194	122
	PrecMin	507	2	9	9	8	11	0	0	12	21	24	10	28
	PrecStd	108	28	19	21	18	24	29	46	53	37	30	36	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.8	8.3	13.0	18.9	22.1	25.7	27.8	26.7	22.1	16.8	13.1	9.7
		MCWB	7.9	6.5	8.8	11.7	15.7	17.9	20.2	18.5	17.2	13.9	12.3	8.9
	2%	DB	7.2	7.1	10.5	15.6	20.1	23.4	26.1	24.6	20.3	15.5	11.7	8.8
		MCWB	6.3	5.8	7.1	10.1	14.8	17.0	19.1	17.7	15.7	13.5	10.7	8.0
	5%	DB	6.2	6.0	8.5	13.8	18.3	21.4	24.6	23.3	18.8	14.5	10.6	8.0
		MCWB	5.3	4.9	6.1	9.0	13.4	15.8	18.2	17.3	15.2	12.7	9.7	7.1
	10%	DB	5.3	4.9	7.0	12.0	16.5	19.9	23.1	21.8	17.5	13.5	9.7	6.9
		MCWB	4.4	3.9	5.3	8.1	12.2	14.9	17.5	17.1	14.4	11.9	8.9	6.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	7.2	9.4	12.2	17.4	19.8	21.6	20.7	18.3	15.2	12.6	9.3
		MCDB	8.7	7.6	12.1	17.6	20.6	23.8	26.3	24.8	21.0	16.0	13.1	9.8
	2%	WB	6.5	6.0	7.5	10.7	15.6	17.6	20.1	19.3	16.7	14.0	11.1	8.3
		MCDB	7.0	6.8	9.3	14.4	18.5	21.6	24.3	22.2	19.0	15.0	11.5	8.7
	5%	WB	5.4	5.0	6.4	9.6	14.3	16.4	19.1	18.5	15.9	13.0	9.9	7.2
		MCDB	6.1	5.9	8.0	12.9	17.0	20.1	23.0	21.2	17.7	14.0	10.3	7.8
	10%	WB	4.5	3.9	5.5	8.5	12.7	15.5	18.2	17.7	15.0	12.2	9.0	6.1
		MCDB	5.2	4.9	6.9	11.4	15.8	19.0	21.9	20.4	16.9	13.3	9.7	6.7
Mean Daily Temperature Range	5% DB	MDBR	3.7	3.7	5.8	7.9	8.3	8.8	8.5	7.9	7.0	5.4	3.8	3.6
		MCDBR	3.6	4.0	8.2	10.6	10.9	11.5	11.4	9.9	8.5	5.8	3.8	3.6
		MCWBR	3.5	3.3	5.3	6.3	6.5	6.2	5.6	4.5	4.8	4.4	3.5	3.7
	5% WB	MCDBR	3.6	3.9	7.3	9.3	9.6	10.5	9.7	8.4	7.1	5.8	3.5	3.4
		MCWBR	3.6	3.4	5.2	6.0	6.4	6.3	5.4	4.8	4.6	4.7	3.4	3.5
Clear-Sky Solar Irradiance	taub	0.315	0.338	0.360	0.394	0.391	0.385	0.406	0.407	0.375	0.359	0.339	0.307	
	taud	2.292	2.313	2.304	2.278	2.344	2.388	2.347	2.351	2.420	2.398	2.334	2.269	
	Ebn at Noon	599	728	803	825	852	861	832	804	783	696	563	512	
	Edh at Noon	55	77	98	116	115	111	114	107	88	71	54	45	
All-Sky Solar Radiation	RadAvg	0.45	1.06	2.44	4.23	5.46	5.83	5.39	4.34	3.07	1.50	0.58	0.30	
	RadStd	0.09	0.17	0.27	0.49	0.54	0.43	0.64	0.31	0.29	0.22	0.10	0.03	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (59 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	-161	N/A	N/A

Nomenclature: See separate page